MAHATMAGANDHIUNIVERSITY,KOTTAYAM

MGU-UGP (HONOURS) SECOND SEMESTER EXAMINATION

(2024ADMISIONONWARDS)

MG2DSCZGY101 – Biological Basis of BehaviourII

Model Question Paper

Duration:1.5hrs MaximumMarks:50

Studentsshouldattemptatleastonequestionfromeachcourseoutcometoenhancetheiroverall outcome attainability.

Part A

Fill in the blanks

Questions Answer **All**

Questions

Eachquestioncarries**1**mark

1. The bacterial cell wall is composed of -------

a) Peptidoglycan b) Chitin c) glucose d) Cellulose [Remember][CO: 1]

2. In which type of signalling do cells secrete molecules that bind to their own receptors?

a)juxtracrine b) paracrine c) endocrine d) autocrine [Understand][CO: 1]

3. DNA synthesis occurs during the _____phase of cell cycle

a) G1 b) G2 c) S d) M [Remember][CO: 1]

4. ------represents alternatives of a character and are present on two separate chromosomes of a homologous pair

a) synapse b) chromatid c) allele d) gene [Remember][CO: 2]

5. The dihybrid phenotypic ratio

a) 1:3:1 b) 1:2:1 c) 9:3:3:1 d) 9:7 [Understand][CO: 2]

6. Which enzyme is responsible for the conversion DNA to mRNA

a) . DNA polymerase b) RNA polymerase c) ligase d) helicase [Understand][CO: 2]

7. Example for sex limited character

a) Brachydactyly b) baldness c) cleft lip d)haemophilia [Remember][CO: 3]

8. The physical appearance of an organism is ----

a) Phenotype b) genotype c) allele d) heterogamete [Remember][CO: 3]

9. -----is a condition where an organism have more than 2 complete sets of chromosomes

a) polyploidy b) diploid c) haploid d) aneuploidy [Understand][CO: 3]

10.The number of chromosomes in human ------- karyotype

a) 54 b) 42 c) 46 d) 38 [Remember][CO: 4]

[1x10=10]

Part**B**

ShortAnswerType Questions

Answer **10**Questions

Eachquestioncarries**2**marks

11. Differentiate prokaryote and eukaryote. [Understand][CO: 1]
12. List the substages of prophase I of meiosis I [Understand][CO: 1]

13. Define cell communication [Understand][CO: 1]
14. Comment on basic principles of cell communication [Understand][CO: 1]
15. Explain monohybrid cross with example. [Understand][CO: 2]
16. A farmer has a tall pea plant but is unsure whether it is homozygous (TT) or heterozygous (Tt) for the tallness trait. Suggest a genetic method to determine its genotype and explain the expected outcomes. [Apply][CO: 2]
17. Describe central dogma in molecular biology [Understand][CO: 2]
18. Differentiate dominant and recessive mutation. [Understand][CO: 3]
19. A newborn is diagnosed with trisomy 21. Identify the disorder and describe its clinical manifestations. [Apply][CO: 3]
20. What are the ethical issues related to genetic experiments in human. [Analyse][CO: 4]
21. How can pedigree analysis be used to identify patterns of inheritance and genetic disorders? [Analyse][CO: 4]
22. Differentiate euthenics and eugenics [Understand][CO: 4]

[2x10=20]

Part**C**

Short EssayTypeQuestions Answer **5** Questions

Eachquestioncarries**4**marks

23. Describe different stages of mitosis. [Understand][CO: 1]
24. Describe different types of cellsignallingmolecules [Understand][CO: 1]
25. What are the benefits and risks of genetic engineering, and how do they influence human health and the environment? [Analyse][CO: 4]
26. Explain sex linked inheritance with an example. [Understand][CO: 2]
27. Describe the structure of DNA. [Understand][CO: 2]
28. Explain different types of gene mutation [Understand][CO: 3]
29. Describe sex chromosomal abnormalities with example. [Understand][CO: 3]

[4x5=20]

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